Extracting plant stress genes and functions from scientific literature to support annotation of switchgrass (*Panicum virgatum L.*) genes

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Abstract: Plants are frequently exposed to a plethora of environmental stresses, affecting and limiting crop yield worldwide. These environmental stresses can be biotic (e.g., bacteria, viruses, fungi, and insects) or abiotic (e.g., drought, extreme temperatures, and salinity). Plants produce a wide variety of responses, for example, a change in the rate of photosynthesis and stomatal closure as they learn to endure these stresses. Crop improvement programs have focused on developing crops that can tolerate these stresses; thus, the number of publications on plant stress has significantly increased over the years. To help annotate Switchgrass genes involved in biotic and abiotic stress, we have established a pipeline that integrates text mining methods to efficiently retrieve plant stress genes from scientific literature and link them to their function. We used information from PudMed API from E-Utilities, Textpresso, pGenN, UniProt, and Europe PMC annotations for gene ontology and GeneRIF. The current collection, stored in MongoDB, includes 2,766 abstracts, 3,716 unique plant stress-responsive genes, 861 GO terms and 1,007 functional annotation GeneRIF sentences. We used this information to predict the role of stress-responsive genes in switchgrass, a bioenergy-relevant crop, using the collection of publications in this study to search for evidence described in other plants under stress. An additional outcome of this work is a set of publications for UniProt entries with annotations that can be submitted to the UniProt Knowledgebase via the community submission system. We expect that these annotated plant stress genes will be helpful for further understanding the mechanisms of stress tolerance in plants.

INTRODUCTION

Switchgrass is a promising biofuel crop; however, environmental stresses such as drought and heat affect switchgrass limiting its biofuel production. Approaches to improve crop tolerance to stresses require adequate knowledge of the underlying genes, primarily explored in a few model plant species (Marrow III et al. 2014, Lovell et al. 2021). Thus, this work presents a resource to provide additional annotation to genes in switchgrass and plants in general.

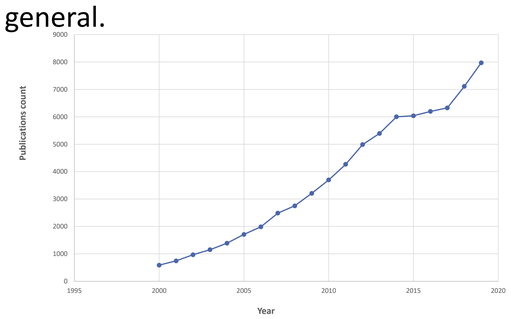


Fig. 1 Number of scientific publications on stress studies in plants from PUBMED database from 2000 to 2020

PudMed database was queried using this script (pubmed - ((“2000/MM01/01 "[Date - Publication] : “2018/12/31"[Date - Publication])) AND stress) AND "Plants"[MeSH]). The dates for the beginning and end of each year were inserted to retrieve the literature for each year (data was retrieved from PudMed in April 2021)

OBJECTIVES

i. Establish a pipeline to efficiently retrieve plant stress genes and their function

ii. To predict the role of stress-responsive genes in switchgrass

ii. To connect resources to understand plant stress

MATERIALS AND METHODS

- **Textpresso central**: TM system that allows for literature search using keywords and by category using controlled vocabulary
- **PgenN**: Automatically identifies plant gene names in abstracts and link them to a database entry
- **PudMed E-Utilities API:** Allows the search and retrieval of specific data from NCBI with controlled search fields
- **Europe PMC:** ePMC offer annotations based on TM approaches to retrieve comprehensive annotations from publications (functional annotations and GO terms)

Workflow to extract plant stress genes and function from literature

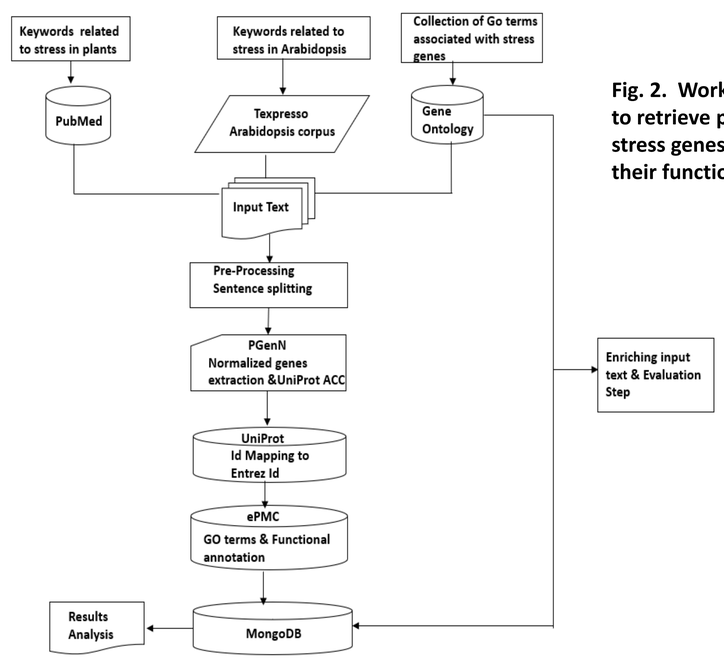


Fig. 2. Workflow to retrieve plant stress genes and their function

RESULTS

Statistics of dataset collected before text pre-processing(Table1) and data stored in MongoDB (Table 2)

Table 1.

No. of Abstracts	Keywords	Database/Resource
8,227	Abiotic AND Stress	PudMed
3,533	Biotic AND Stress	PudMed
4,691	GO terms related to stress	Gene Ontology
428	Abiotic/biotic AND stimulus from Arabidopsis corpus	Textpresso
Abstracts = 16,879	Sentence splitter used to split the abstracts	
Duplicates = 2,820		
Total abstracts processed= 14, 059		

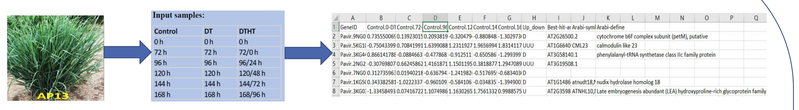
Table 2.

Name (No. of)	(MongoDB)
Total abstracts	2,766
Abstracts with functional annotation sentences	1,007
Abstracts with stress genes and GO terms	1,759
Unique plant stress genes	3,716
Unique GO terms	861
Plant species	162

MongoDB collection

Fig. 3 Example of a document stored in the MongoDB collection

Transcriptome data on switchgrass during stress



Hayford et al. 2021 BMC Plant Biology (submitted-under review)

Example of predicted annotations for switchgrass genes

Table 3.

DEGs from in-house RNA-Seq data	Gene expression	Ortholog	Gene enrichment analysis(AgriGO) OR TAIR annotation Biological process	Evidence from the MongoDB collection	Stress gene/UniProt accession	% Identity of ortholog to *Panicum Virgatum* gene	Annotation to infer-information from our	Evidence identified In GO
Pavir.1KG386300.v4.1	Downregulated DTHT 96/24h and DTHT 168/96h	AT2G37040.1 ATPAL1,PAL1 PHE ammonia lyase 1	No significant term from AgriGO TAIR: Cinnamic acid biosynthetic process Defense response Drought recovery Lignin catabolic process L-phenylalanine catabolic process Pollen development Response to Karrikin Response to oxidative stress Response to UV-B Salicylic acid catabolic process	PMID: 27439459 (*Arabidopsis thaliana*) PMID:26389875 (*Solenostemon scutellarioides*)	PAL1 P35510 PAL1 L0BBX7	75.2% E-value: 0.0 77.5%	Cold acclimation (GO:0009631(P)) Secondary metabolism (GO:0019748(P)) Anthocyanin synthesis (GO:0009718(P)) (GO:0009058(P))	No No No
Pavir.1KG530100.v4.1	Upregulated DT 168/96 h	AT2G18700.1 ATTPS11,ATTPSB,TPS11 trehalose phosphatase/synthase 11	AgriGO: single-organism biosynthetic process (GO:0044711(P)	PMID: 23430324 (*Zea mays*)	TPS11 Q1EG72	61.7% E-value: 0.0	Defense response (GO:0006952(P)) , Phytoalexin biosynthesis (GO:0052315(P)) , Ethylene biosynthesis (GO:0009693(P))	No
Pavir.5KG421100.v4.1	Upregulated DT 120/48 h	AT4G18170.1 ATWRKY28,WRKY28 WRKY DNA-binding protein 28	AgriGO: organic cyclic compound biosynthetic process (GO:1901362) (P) TAIR: positive regulation of leaf senescence Regulation of cell fate specification	PMID: 23749099 (*Arabidopsis thaliana*)	WRKY28 Q8VVW2	86.9% E-value:4.46754E-47	Oxidative burst (GO:0045730 (P)), hyphal growth (GO:0030448 (P)	No

A set of publications for UniProt entries with annotations

Table 4.

UniProt AC	UniProt Status	Protein Name	Organism	PMID	Article Title	Category	Annotation	ORCID
L0BXX7	Unreviewed	Phenylalanine ammonia-lyase	Plectranthus scutellarioides (Coleus) (Solenostemon scutellarioides).	26389875	Isolation and Functional Characterization of a Phenylalanine Ammonia-Lyase Gene (SsPAL1) from Coleus (Solenostemon scutellarioides (L.) Codd).	[Function][Subcellular location][PTM / Processing] [Expression]	**Protein/gene_name:** PAL1. **Function:** PAL1 shows high catalytic activity to convert L-phenylalanine to trans-cinnamic acid. **Comments:** It is found in the cytosol. Its gene expression is enhanced by light.	0000-0002-8583-9432
A0A216U1U9	Unreviewed	Em-like protein	Lactuca sativa (Garden lettuce).	29417232	A new Em-like protein from Lactuca sativa, LsEm1, enhances drought and salt stress tolerance in Escherichia and rice.	[Function][Subcellular location][Expression]	**Protein/gene_name:** LsEm1. **Function:** LsEm1 is a stress-responsive gene involved in drought and salt tolerance. Over-expression of LsEm1 in transgenic rice enhances drought and salt tolerance in rice, causing up-regulation of OsCDPK genes. **Comments:** Transcript accumulation of the LsEm1 gene was detected in the leaves and roots during the seedling period, with some high transcripts in all examined tissues during the flowering and mature periods. LsEm1 protein may be in the cytoplasm.	0000-0002-8583-9432
O49195	Reviewed	Vegetative storage protein 1	Arabidopsis thaliana (Mouse-ear cress).	23166664	The crystal structure of Arabidopsis VSP1 reveals the plant class C-like phosphatase structure of the DDDD superfamily of phosphohydrolases.	[Structure]	**Protein/gene_name:** VSP1. **Comments:** VSP1 is a dimer. VSP1 structure possesses a core Rossmanoid domain and a smaller helical domain. PDB:4FYP.	0000-0002-8583-9432

DISCUSSION

- As a case study, we have used various computational approaches to predict the function of the Phenylalanine ammonia-lyase I (PAL1) gene in switchgrass. PAL1 is involved in the phenylpropanoid pathway
- **Further studies:** Enzymatic activity test to confirm gene (PAL1) function

REFFERENCE

Zhu Q, Xie X, Lin H, Sui S, Shen R, Yang Z, Lu K, Li M, Liu Y-G. Isolation and Functional Characterization of a Phenylalanine Ammonia-Lyase Gene (*SsPAL1*) from *Coleus* (*Solenostemon scutellarioides* (L.) Codd). *Molecules*. 2015; 20(9):16833-16851. https://doi.org/10.3390/molecules200916833

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